

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Effect of Formulation and Food on Calderasib (MK-1084) in Healthy Adult Participants (MK-1084-011) **Short Title/Acronym:** MK-1084-011
Protocol Number: MK-1084-011
NCT Number: NCT06942741
Sponsor Name: Merck Sharp & Dohme LLC
Investigational Product: Caldearasib (MK-1084) [oral tablets]
Phase of Development: Phase 1 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * Part 1: To compare the pharmacokinetics of calderasib in the blood when taken as two different types of oral tablets (Formulation Effect). * Part 2: To learn what happens to calderasib in the blood when taken on an empty stomach compared to after a meal (Food Effect).

Secondary Objectives: To evaluate the overall safety and tolerability of calderasib in healthy participants, including assessing the number of participants who discontinue treatment due to adverse events (AEs).

2.2 Background & Rationale Caldearasib (MK-1084) is a selective KRAS G12C inhibitor. Before advancing into broader late-stage oncology trials, it is crucial to determine how the body absorbs and processes different tablet formulations of the drug and to establish whether the medication should be administered with or without food. This study will guide optimal dosing instructions for future clinical studies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basic Science)
Control Type: Active/Crossover **Blinding:** Open-Label
Randomization: Randomized **Interventional Model:** Crossover Assignment

3.2 Planned Enrollment & Duration Target N: 52
Number of Sites: 1 (Celerion, Tempe, Arizona, United States)
Estimated Study Start: 13-May-2025
Estimated Primary Completion: 22-Jun-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- 1. Healthy adult participants.
- 2. Ability to understand and provide informed consent.
- 3. Willingness to comply with all study procedures, including fasting and dietary restrictions during the crossover periods.

4.2 Exclusion Criteria

- 1. Has a history or presence of a clinically significant medical or psychiatric condition or disease.
 - 2. Has a history of cancer (malignancy).
 - 3. Has positive results for human immunodeficiency virus (HIV), hepatitis B surface antigen, or hepatitis C virus.
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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple specific dosages across two study parts (Part 1: Alternate Formulations; Part 2: Fasted vs. Fed state). **Route of Administration:** Oral (Tablets).

Duration of Treatment: Short-term exposure across crossover periods.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications per Phase 1 healthy volunteer guidelines.

Prohibited: Concomitant medications that interfere with KRAS G12C absorption or metabolism, or those that could confound safety assessments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to -1	Screening
2	Day 0	Baseline
3	Day 0	Randomization to Crossover Sequence
4	Day 0	Admission to Clinical Pharmacology Unit
5	Day 0	Interim
6	Day 0	Crossover Periods
7	Day 0	Dosing (Fasted/Fed or Formulation A/B)
8	Day 0	Intensive PK Blood Sampling (Cmax, Tmax, C24, etc.)
9	Day 0	Safety Labs
10	Day 0	End of Study
11	Day 0	Follow-up
12	Day 0	Final Vital Signs, AE Monitoring, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetic (PK) parameters of calderasib, including Maximum Concentration (Cmax), Area Under the Curve up to 24 hours (C24), Time to Maximum Concentration (Tmax), apparent terminal half-life (t1/2), apparent clearance (CL/F), apparent volume of distribution (Vz/F), and lag time (tlag). **Measurement Method:** Serial blood sample collections and bioanalytical laboratory analysis.

7.2 Secondary & Exploratory Endpoints * Occurrence of any untoward medical occurrence (Adverse Events / AEs). * Number of participants who discontinue from the study treatment due to an AE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during in-clinic stays. An AE is defined as any untoward medical occurrence in a clinical study participant, temporally associated with the use of study treatment, whether or not considered related to the study treatment. **Serious Adverse Events (SAEs):** Reported within standard regulatory timelines (typically 24 hours). **Data Monitoring Committee (DMC):** Internal safety review committee typical for Phase 1 healthy volunteer studies.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set; Safety Analysis Set. **Sample Size Justification:** Powered adequately to detect bioequivalence or significant differences in Cmax and AUC between formulations and fed/fasted states. **Handling of Missing Data:** Standard handling for PK concentration curves (e.g., excluding profiles with missing critical time points for AUC calculation).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during the active dosing and PK sampling windows in the Phase 1 unit. **Audit Trail:** Electronic Data Capture (EDC) system with strict audit trails for PK sample tracking and AE logging.

11. Study Identifiers & Governance

Primary ID: MK-1084-011 **Registry Number:** NCT06942741 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study to Evaluate the Effect of Formulation and Food on Caldasiasib (MK-1084) in Healthy Adult Participants (MK-1084-011) **Official Regulatory Title:** A Study to Evaluate the Effect of Formulation and Food on Caldasiasib (MK-1084) in Healthy Adult Participants (MK-1084-011)

12. Clinical Framework (Study Specific)

Primary Condition: Healthy Adult Participants **Diagnosis Threshold:** N/A (Healthy volunteers absent of clinically significant medical conditions, cancer, or infectious diseases) **Medical Methodology:** Phase 1 Pharmacokinetic profiling **Optimization Target:** Formulation optimization and food-effect characterization

13. Study Procedures & Methodology

Management Pathway: Clinical Pharmacology Unit (CPU) standard intensive sampling pathway **Education Protocol (HCP):** Strict adherence to PK sampling intervals and crossover dosing sequences **Education Protocol (Patient):** In-clinic dietary controls, fasting instructions, and symptom reporting **Quality Audit Mechanism:** Bioanalytical lab sample integrity monitoring and clinical AE reporting audit

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 7 weeks **Onsite Visit Cadence:** Admission to the clinic during active crossover dosing periods **Remote Monitoring Frequency:** N/A (in-patient Phase 1 protocol) **Checklist Review Interval:** Daily during in-clinic stays for AE and concomitant medication tracking

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Pending	_____	03-Apr-2026				
Clinical Operations Lead	Pending	_____	03-Apr-2026		Lead		
Statistician	Pending	_____	03-Apr-2026				